

DES Pune University, Pune
School of Engineering and Technology
Department of Computer Engineering and Technology
Program: B. Tech in Computer Science and Engineering

Academic Year: 2025-26

Year: Second Year

Semester: II

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Subject: Python Programming

Assignment No.: 15

Date: 02/02/2026

Lab Assignment: 15

Aim:

Write a program to calculate the simple interest, considering 7% ROI for customers below 60 age and 9% for senior citizens.

Objective:

To calculate simple interest using conditional statements and functions.

Theory:

Simple Interest is calculated using the formula:

$$SI = (P \times R \times T) / 100$$

The rate of interest depends on the age of the customer:

- Below 60 years → 7%
- 60 years and above → 9%

Algorithm:

- Start
- Define a function to calculate simple interest
- Read principal, time, and age
- Assign rate based on age
- Calculate simple interest
- Display the result
- Stop

Pseudo Code:

```
FUNCTION calculate_si(p, t, age)
```

```
IF age < 60
```

```
    rate = 7
```

```
ELSE
```

```
    rate = 9
```

```
END IF
```

```
SI = (p * rate * t) / 100
```

```
RETURN SI
```

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END FUNCTION

Code:

```
def calculate_si(principal, time, age):  
    if age < 60:  
        rate = 7;  
    else:  
        rate = 9;  
    simple_interest = (principal * rate * time) / 100;  
    return simple_interest;  
p = float(input("Enter principal amount: "));  
t = float(input("Enter time (in years): "));  
age = int(input("Enter age: "));  
si = calculate_si(p, t, age);  
print("Simple Interest is:", si);
```

Output:

```
Enter principal amount: 10000  
Enter time (in years): 15  
Enter age: 21  
Simple Interest is: 10500.0  
  
Enter principal amount: 10000  
Enter time (in years): 15  
Enter age: 74  
Simple Interest is: 13500.0
```

Conclusion:

The program calculates simple interest correctly based on the customer's age.